AI for Audio

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Overview

Audio

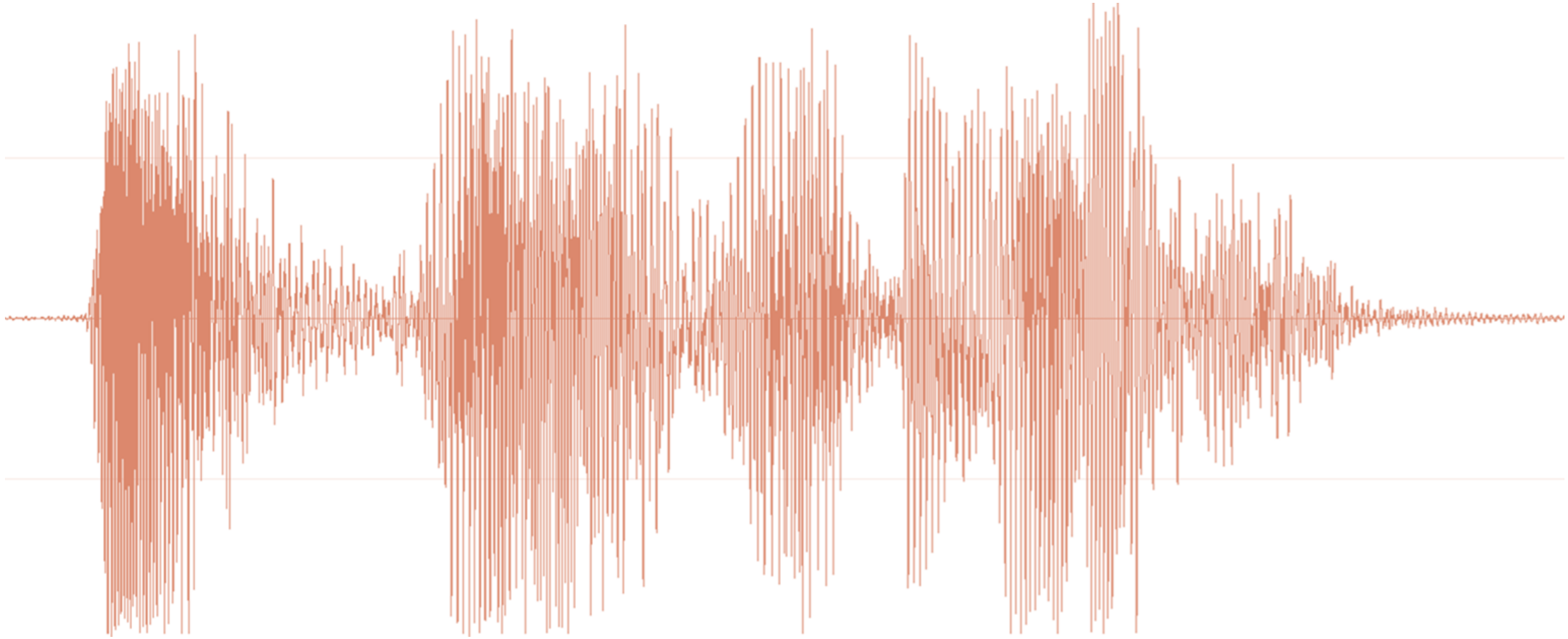
Audio Analysis

Audio Synthesis

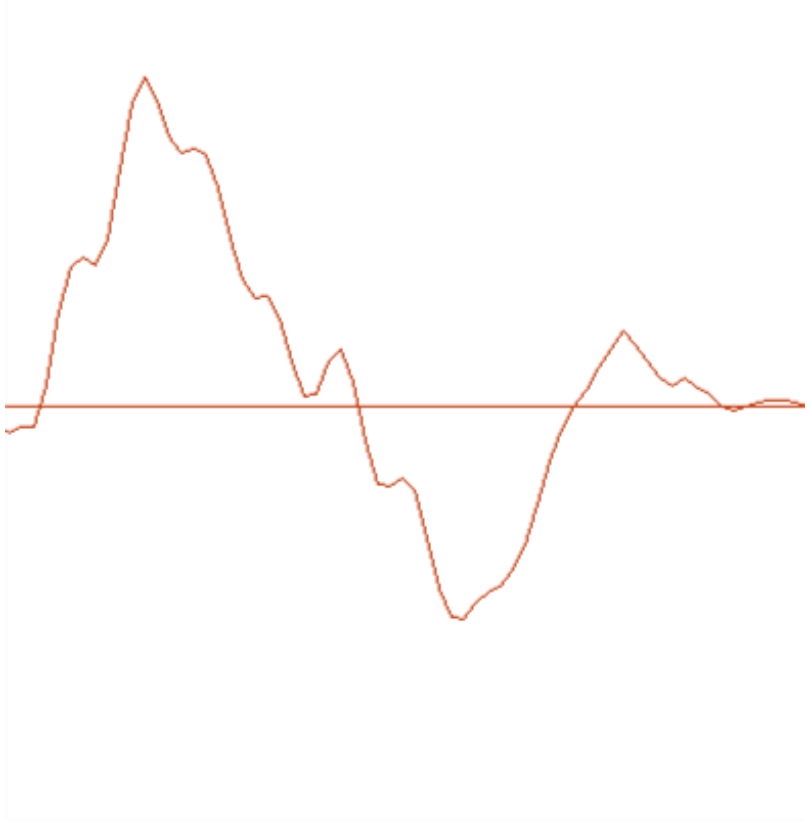
Demo

Audio

Audio



Audio

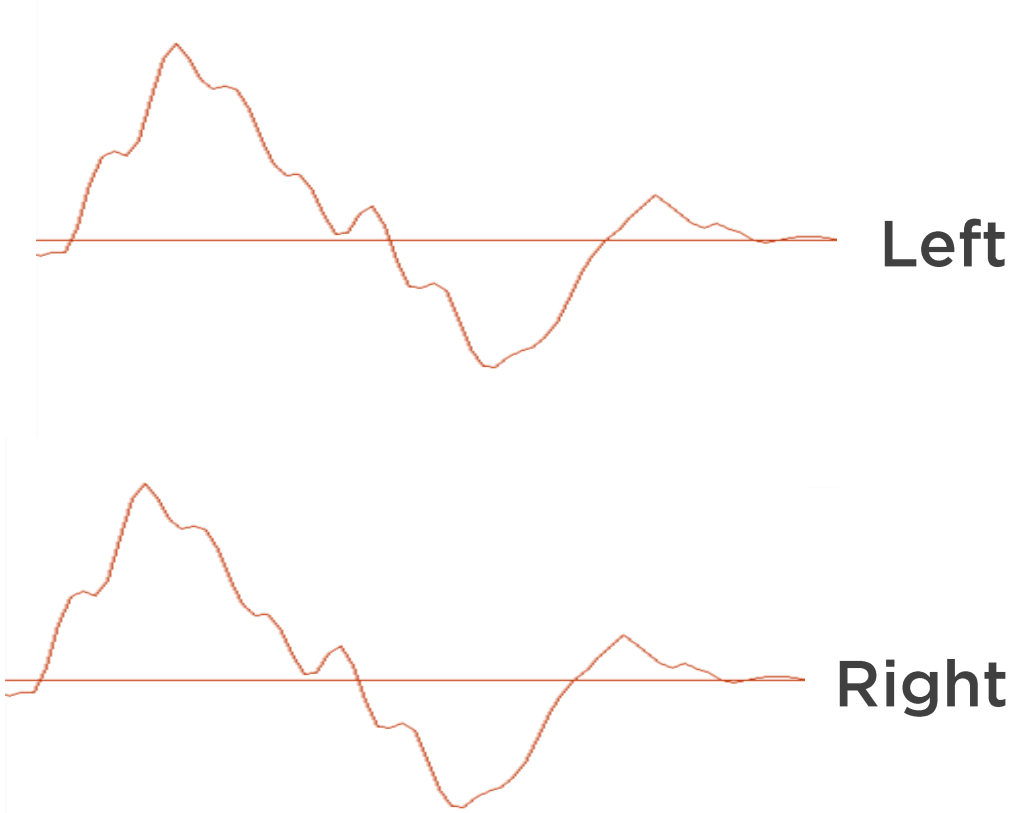


-2	-1	5	10	15	14	17	25	20	...
1	1	0	0	0	0	0	0	0	
1	1	0	0	0	0	0	0	0	
1	1	0	0	0	0	0	0	0	
1	1	0	0	0	0	1	1	1	
1	1	0	1	1	1	0	1	0	
1	1	1	0	1	1	0	0	1	
1	1	0	1	1	1	0	0	0	
0	1	1	0	1	0	1	1	0	

Audio

Audio

Mono



Stereo

Audio Analysis

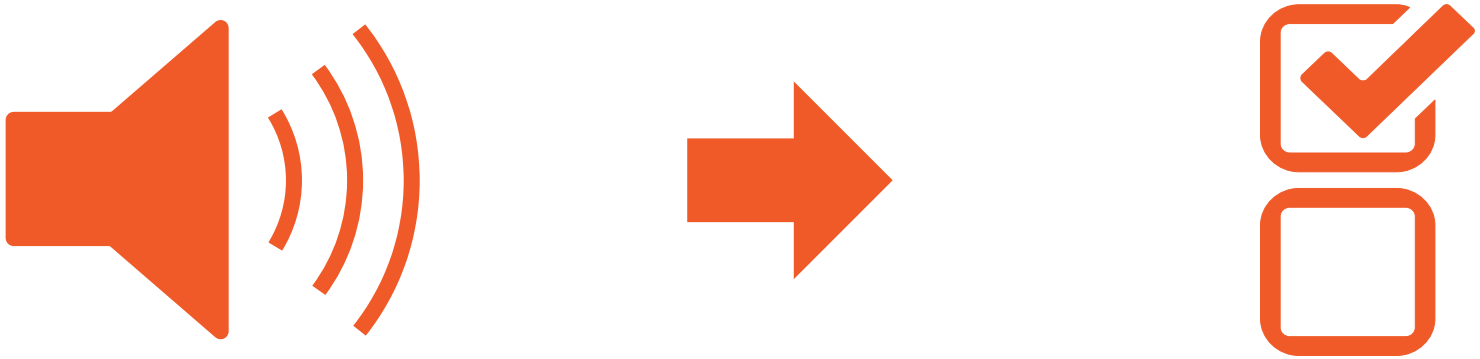
Audio Analysis Tools

Sound classification

Speaker recognition

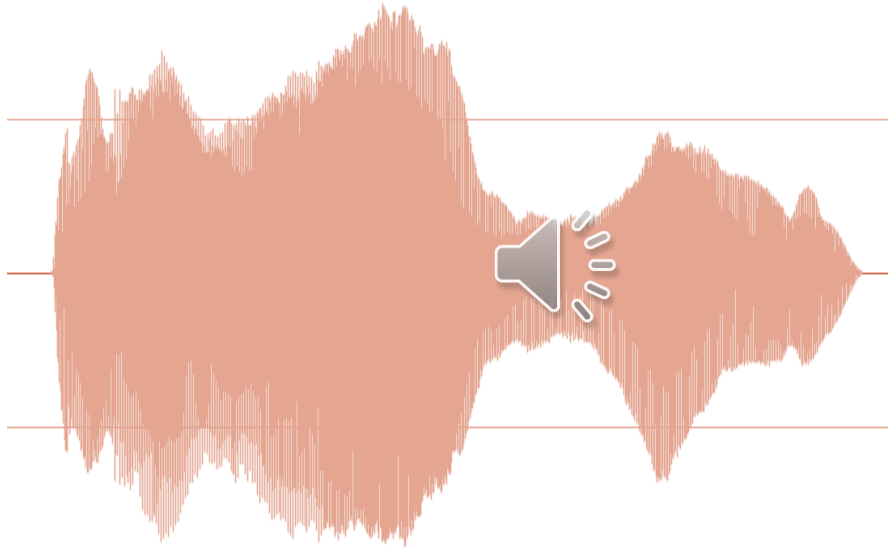
Speech recognition

Sound Classification



Sound Classification

Input



Output

Prediction	Score
Cat	99%
Dog	50%
Bird	10%

Source: Zapsplat.com

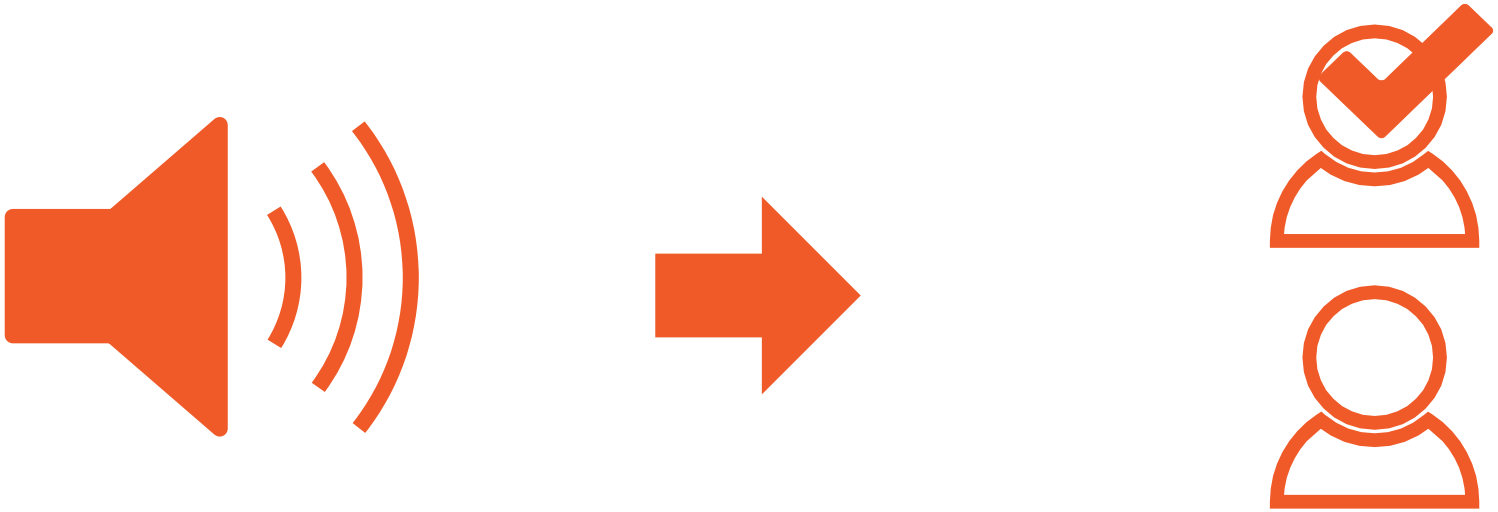
Sound Classification

Gunshot detection

Predictive maintenance

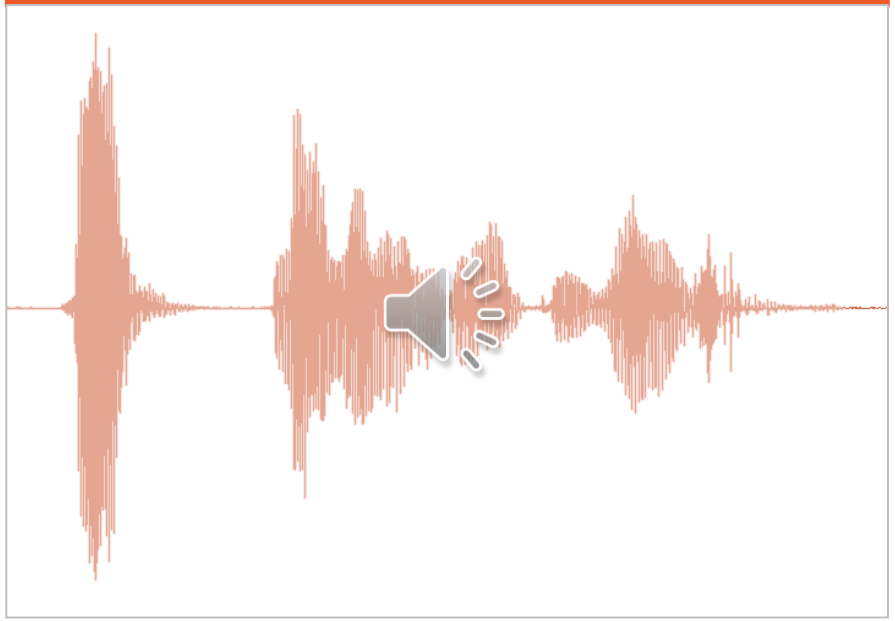
Wildlife conservation

Speaker Recognition



Speaker Recognition

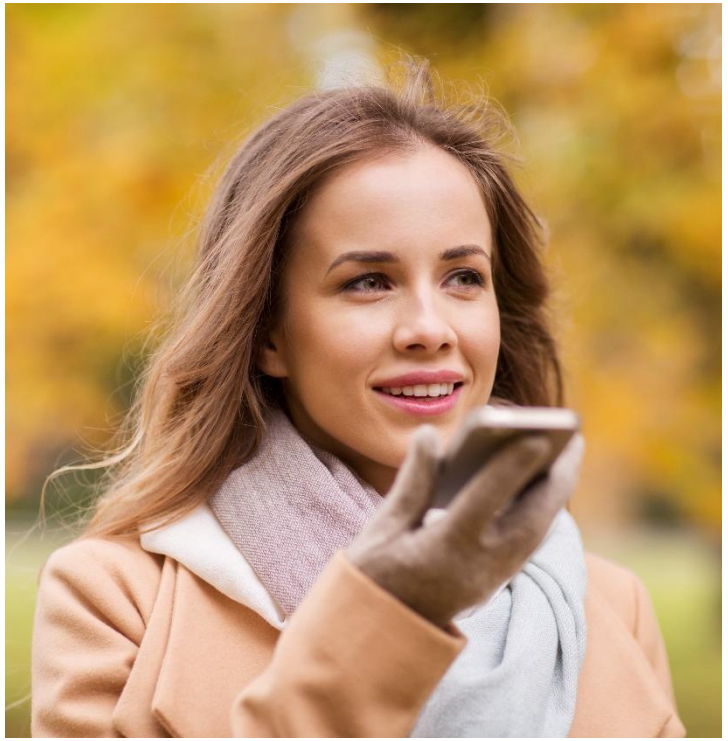
Input



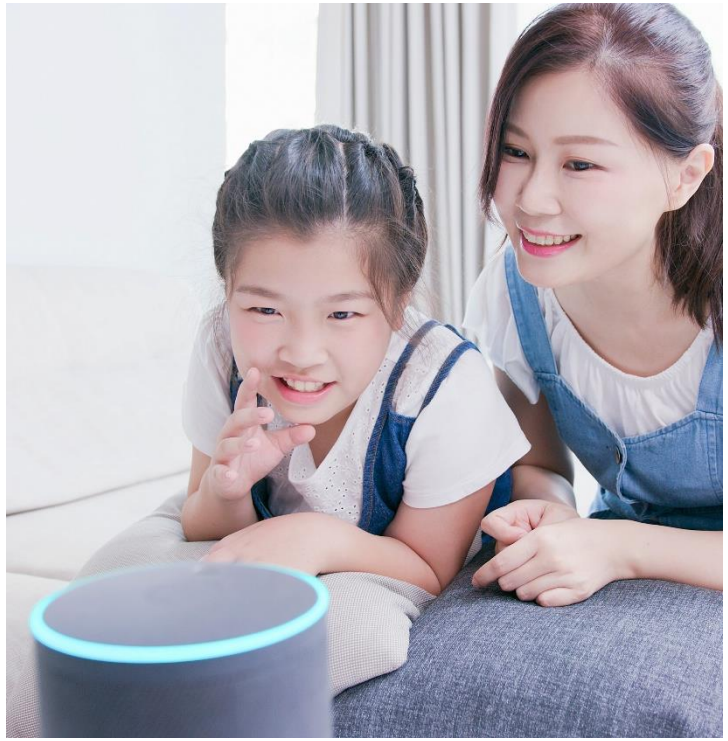
Output

Speaker	Score
Matthew Renze	99%
Zhang San	50%
Priya Singh	10%

Speaker Recognition



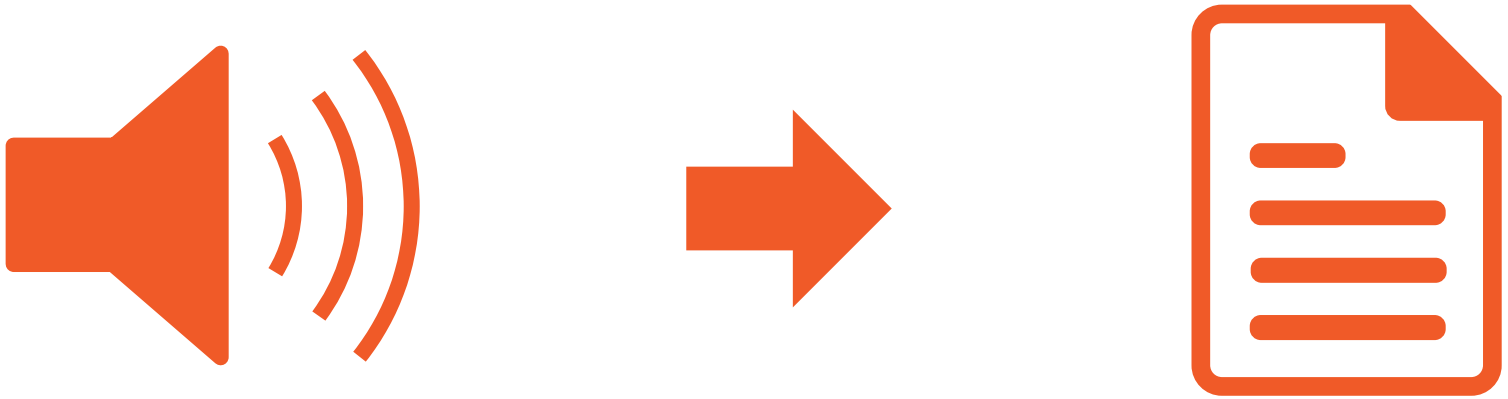
Voice authorization



Speaker personalization

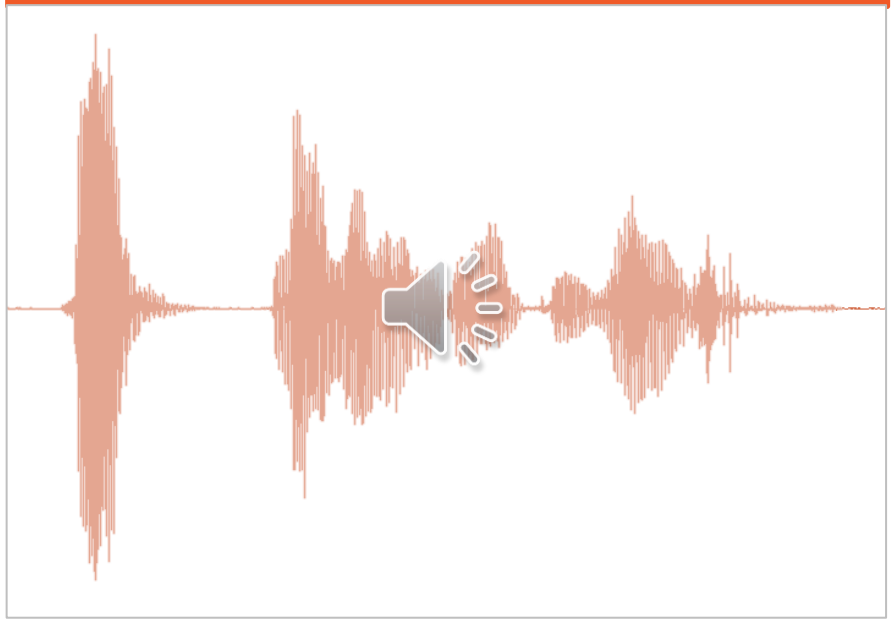
Dialog transcription

Speech Recognition



Speech Recognition

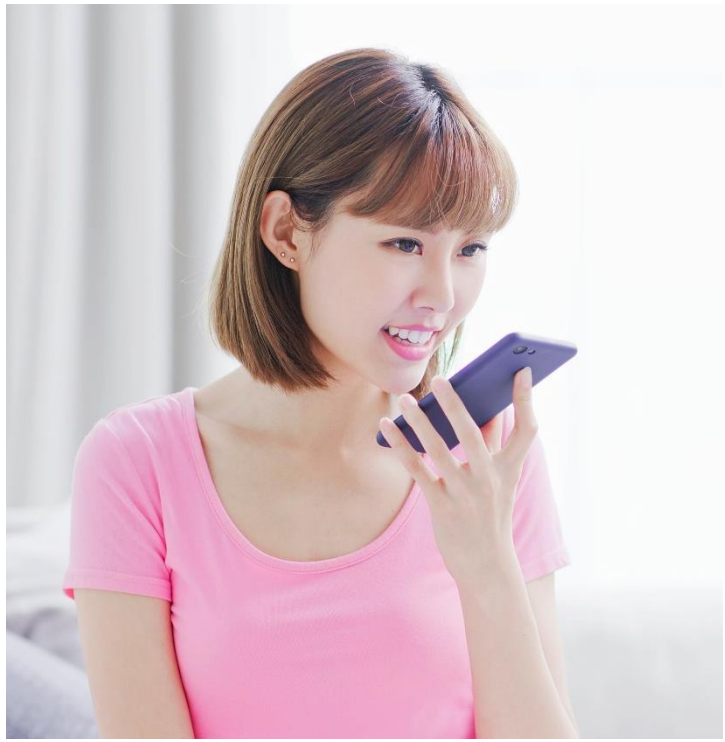
Input



Output

Hi, my name is
Matthew Renze.

Speech Recognition



Voice commands

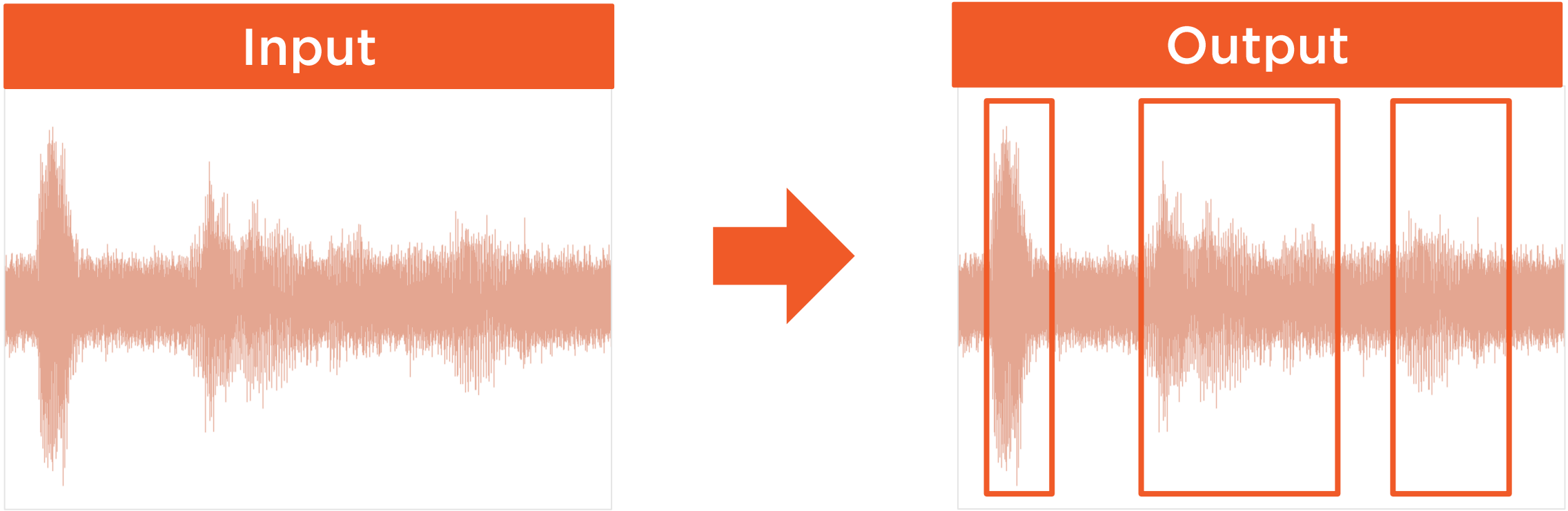
Voice dictation

Closed captions

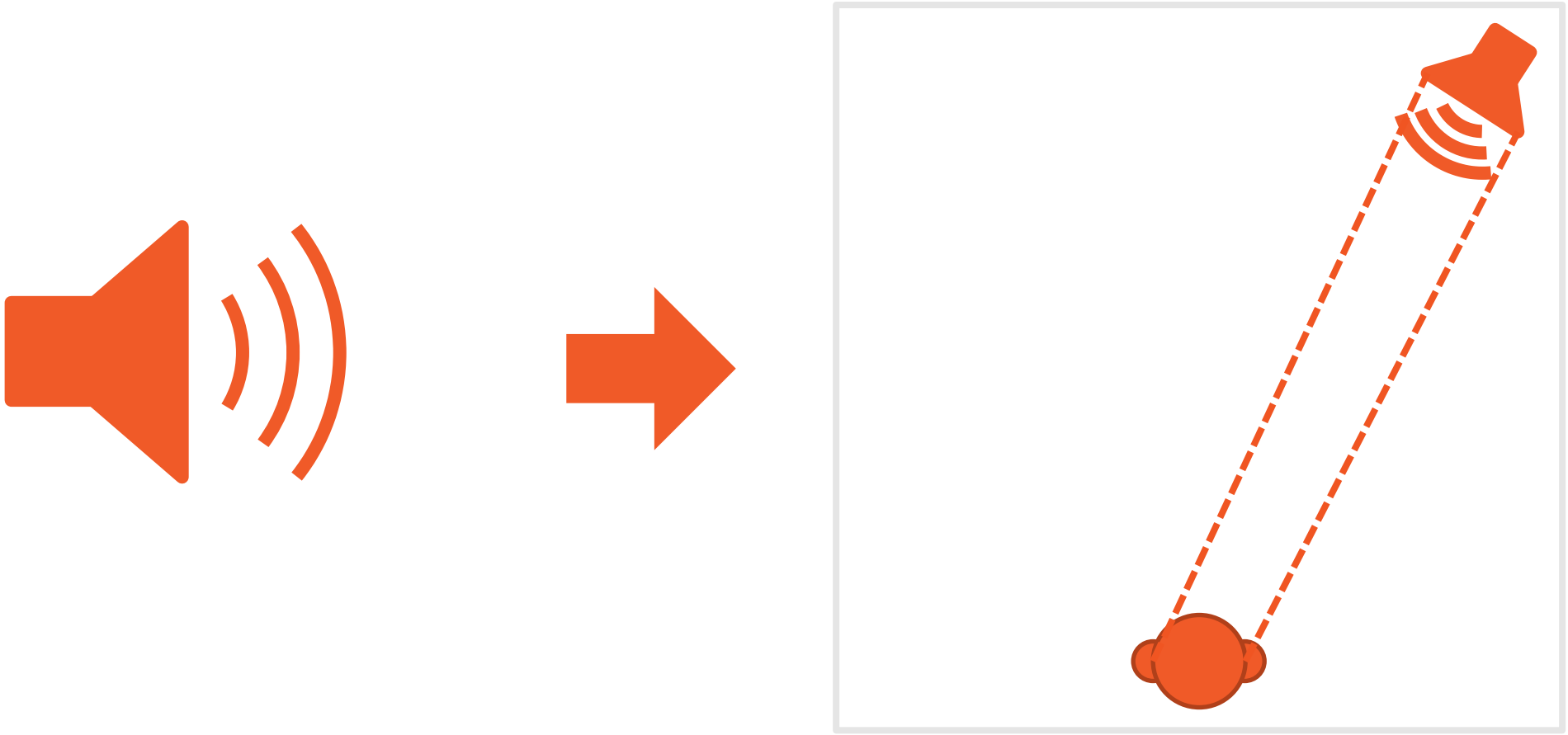
Other Audio Analysis Tasks

Sound classification	Speaker recognition	Speech recognition
Sound detection	Sound localization	Song recognition

Sound Detection

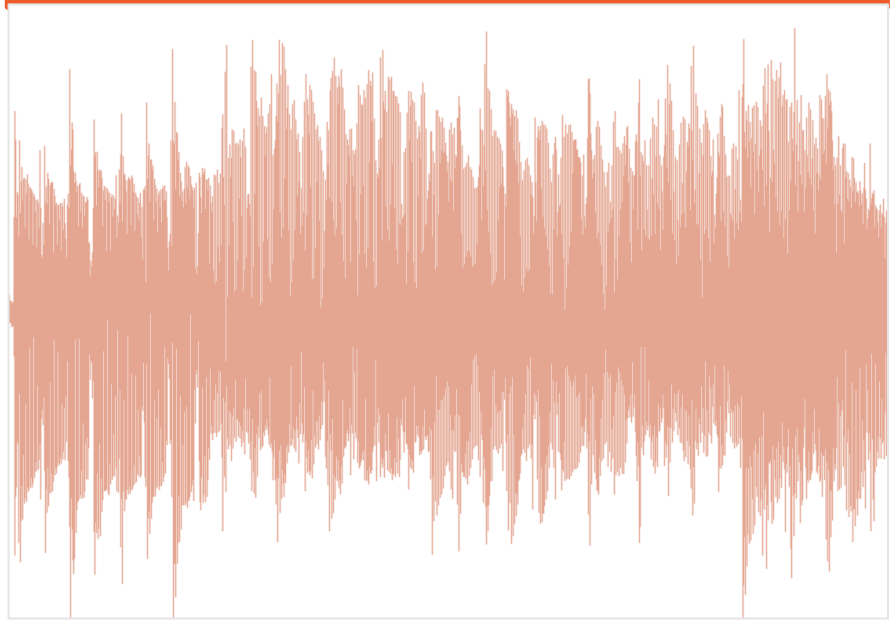


Sound Localization



Song Recognition

Input



Output

Artist	U2
Song	With or Without You
Album	The Joshua Tree
Year	1987

Audio Synthesis

Audio Synthesis Tools

Speech synthesis	Voice synthesis	Speech translation

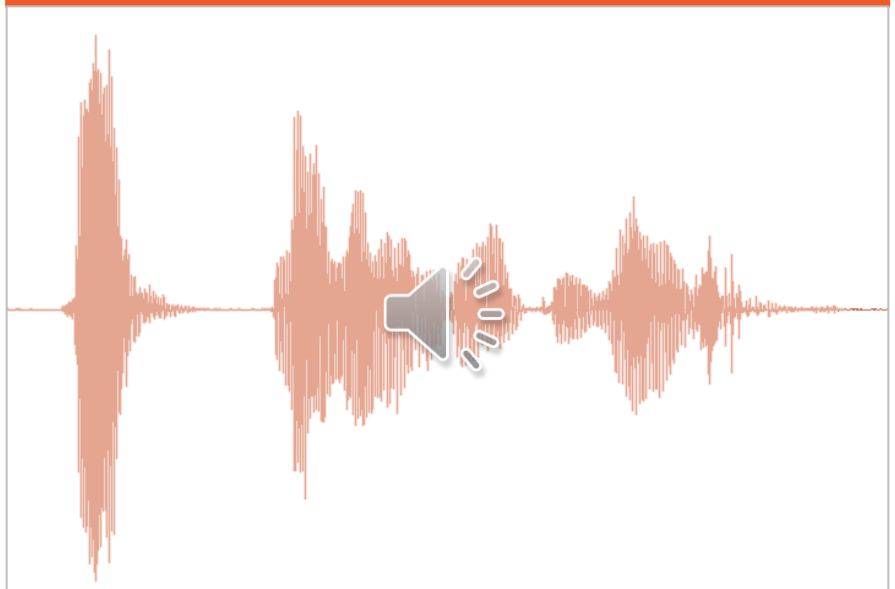
Speech Synthesis

Speech Synthesis

Input

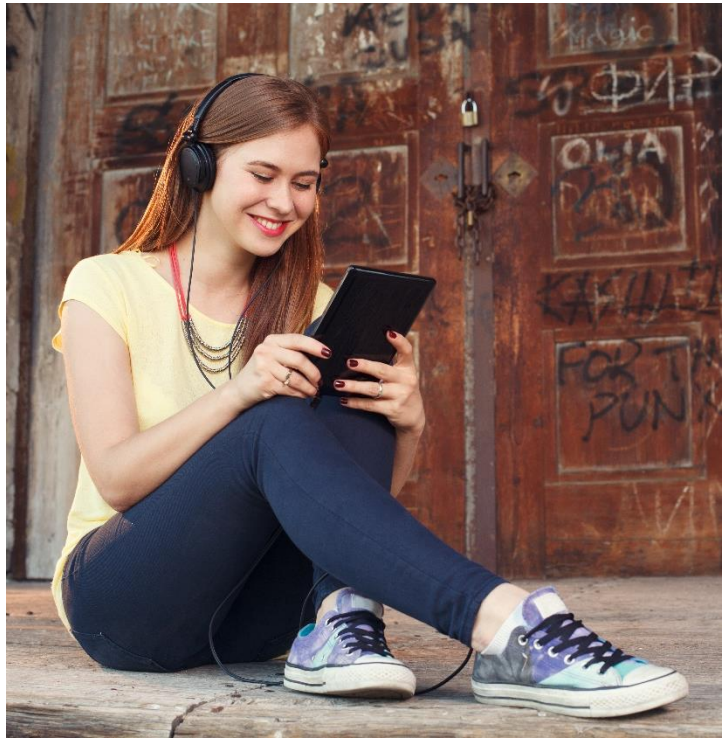
Hi, my name is Matthew Renze.

Output



Speech Synthesis

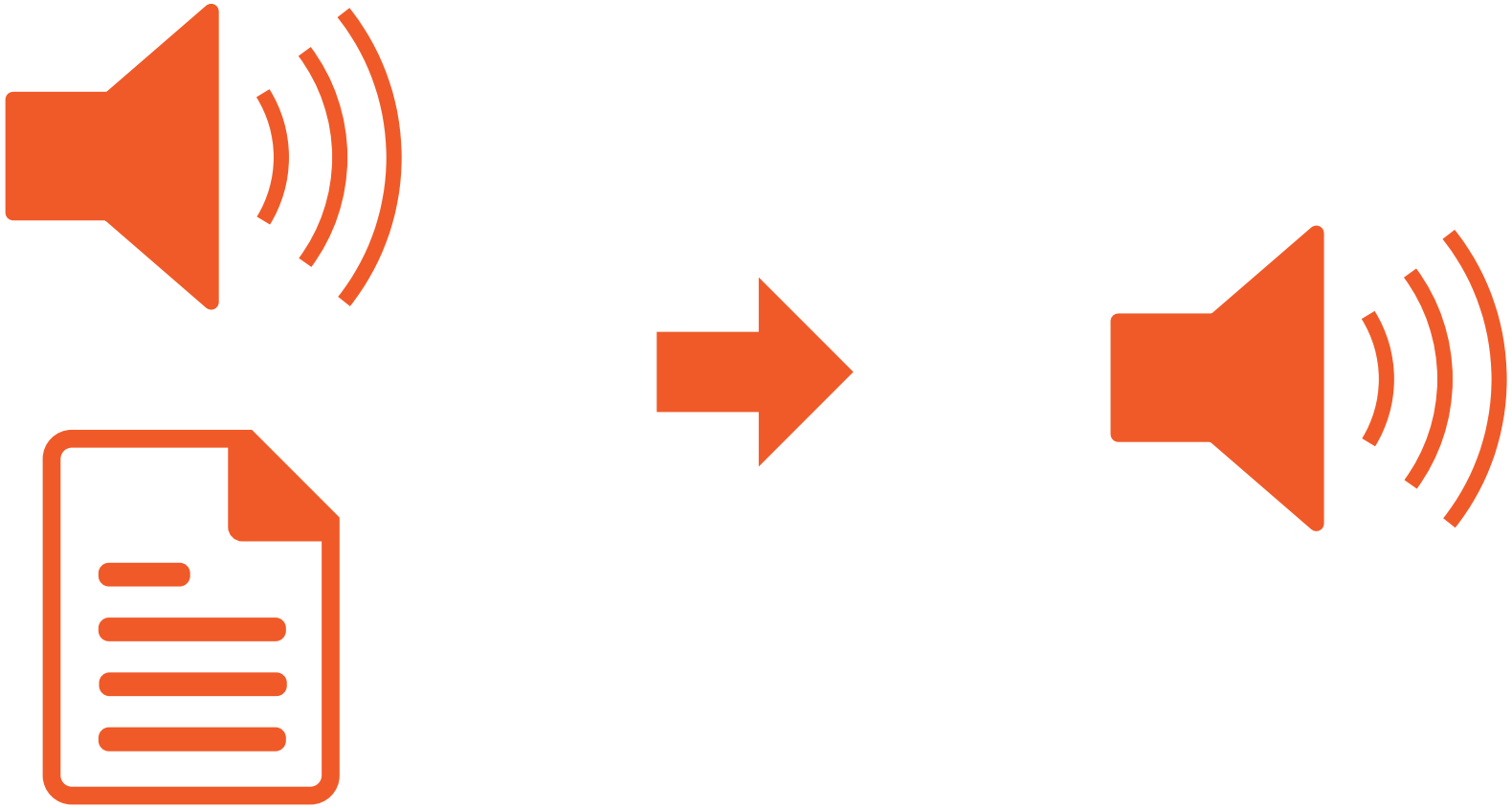
Hands-free apps



Document narration

Natural UI

Voice Synthesis



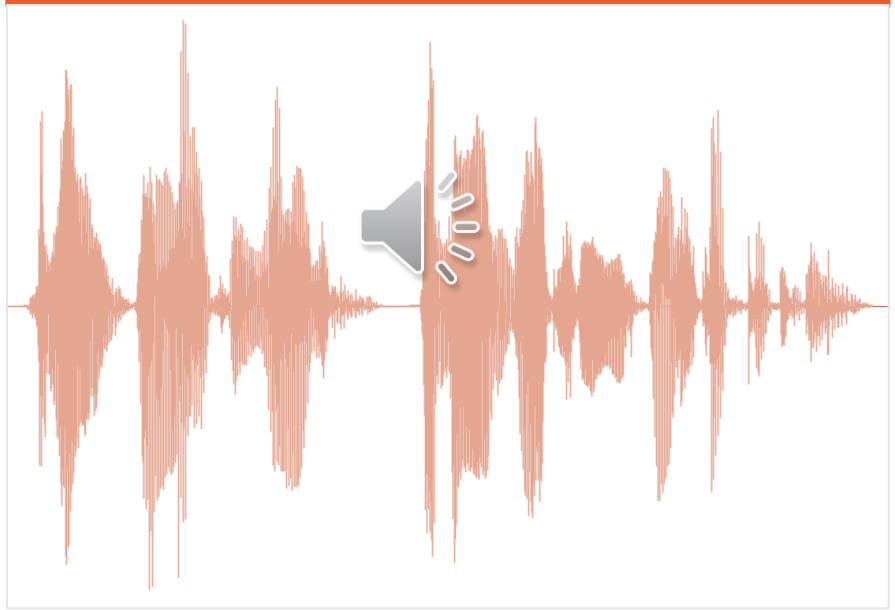
Voice Synthesis

Input

~~Hi, my name is~~ Matthew Renze.

Hello, I'm Matthew Renze.
I'll be your virtual instructor today.

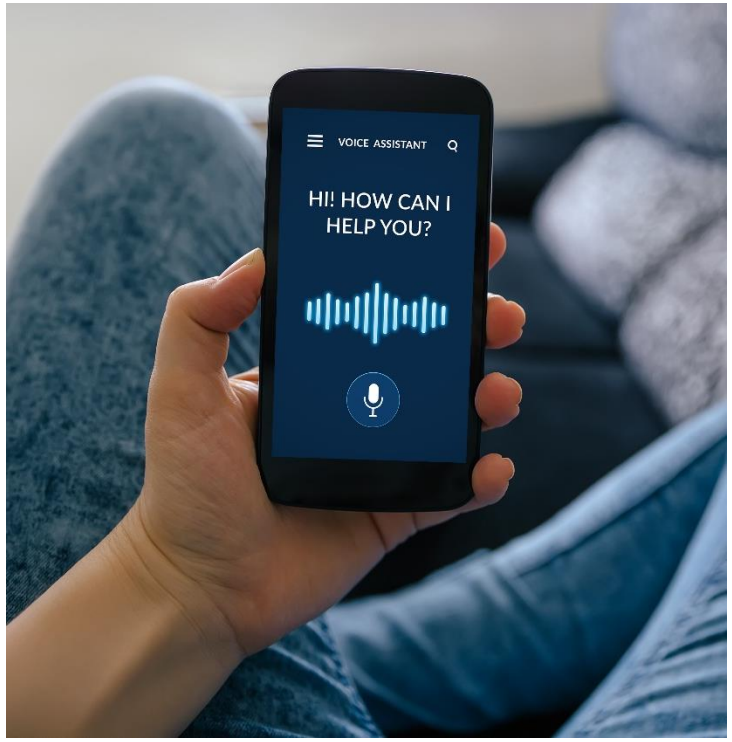
Output



Voice Synthesis

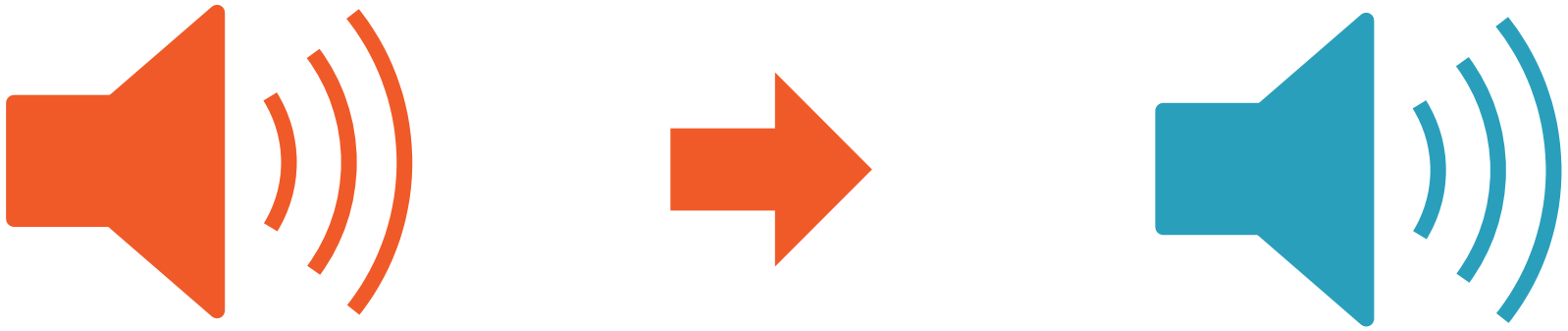
Podcast editing

Audio overdubs



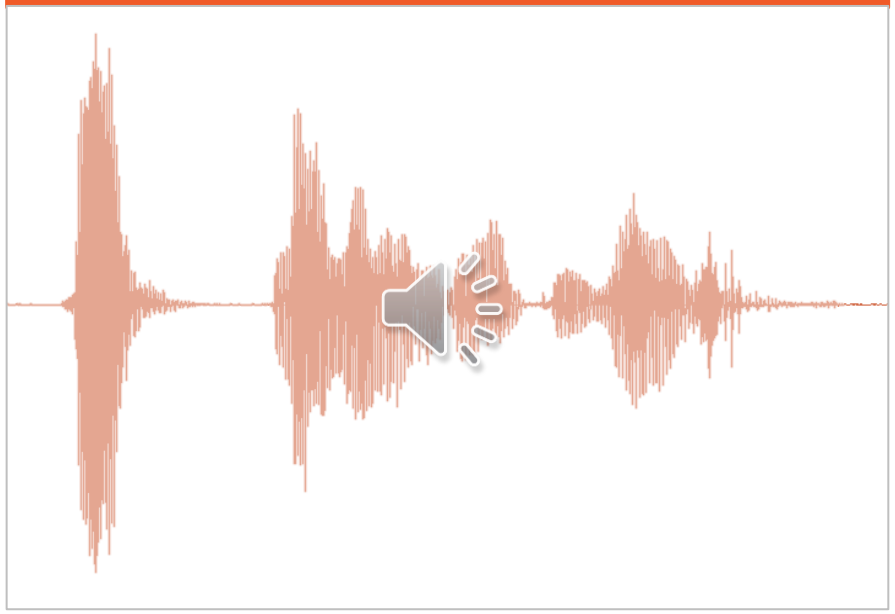
Voice agents

Speech Translation



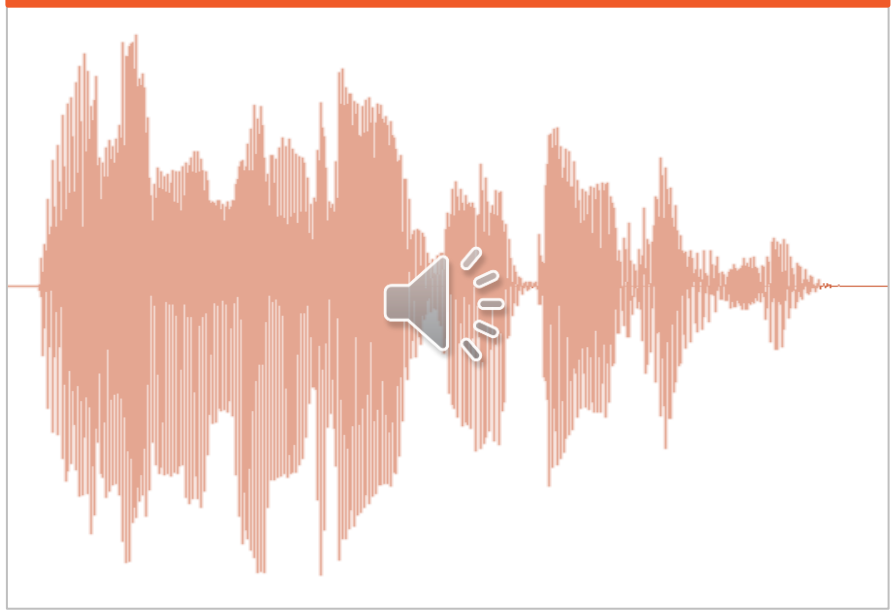
Speech Translation

Input



"Hi, my name is Matthew Renze."

Output



"Hola, mi nombre es Matthew Renze."

Speech Translation

Real-time conversation

Keynote translation

Audio overdubs

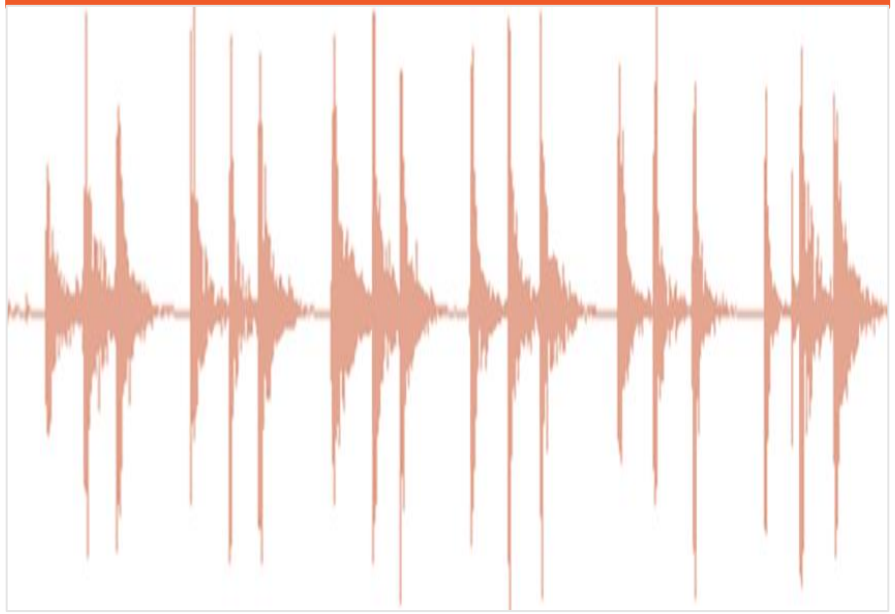
Other Audio Synthesis

Speech synthesis	Voice synthesis	Speech translation
Sound generation	Music generation	Instrument translation

Sound Generation

Input

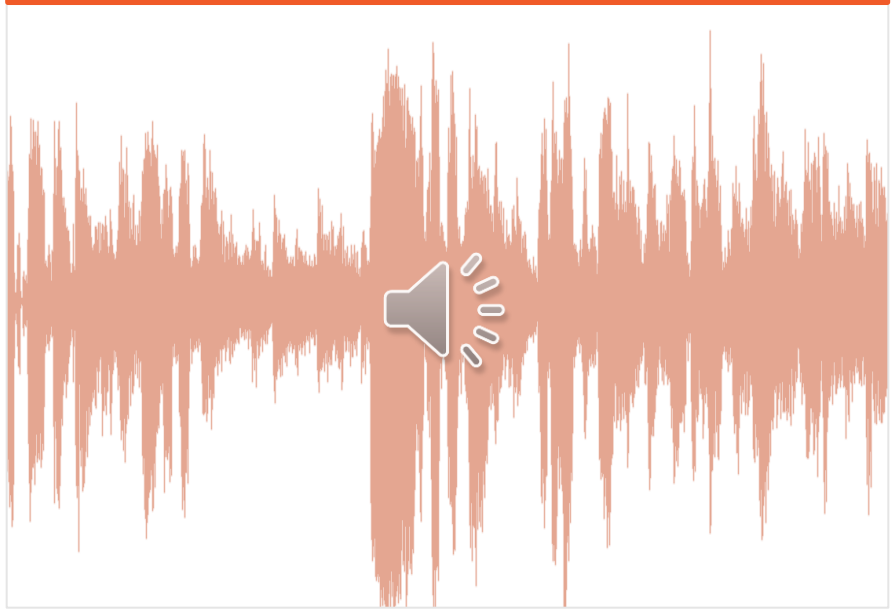
Output



Music Generation

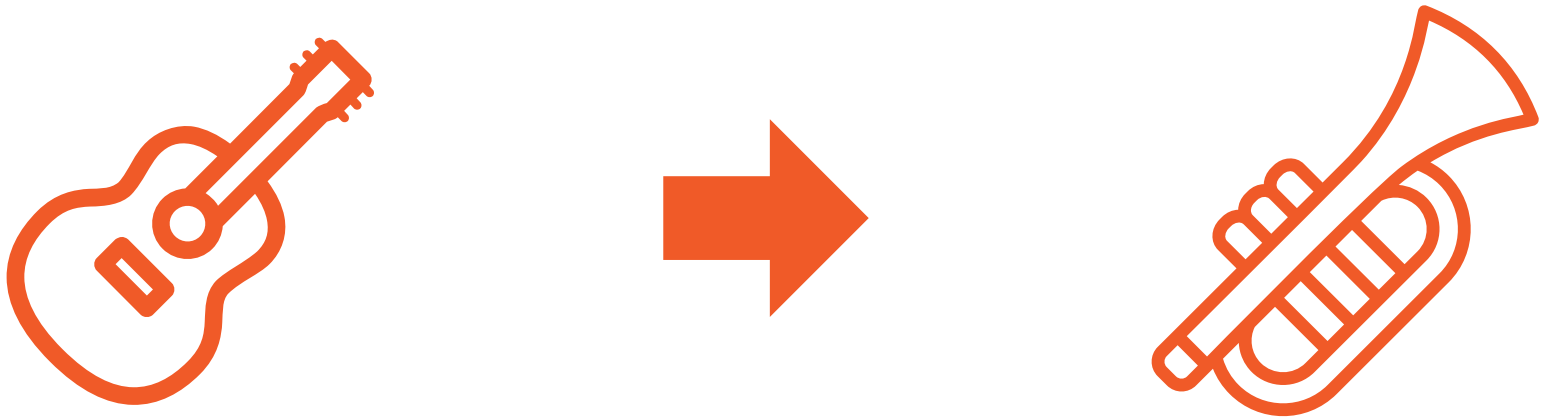
Input	
Genre	Classic Pop
Artist	Frank Sinatra
Lyrics	It's Christmas time and you know what that means. Oh, it's hot tub time as I ...

Output



Source: OpenAI

Instrument Translation

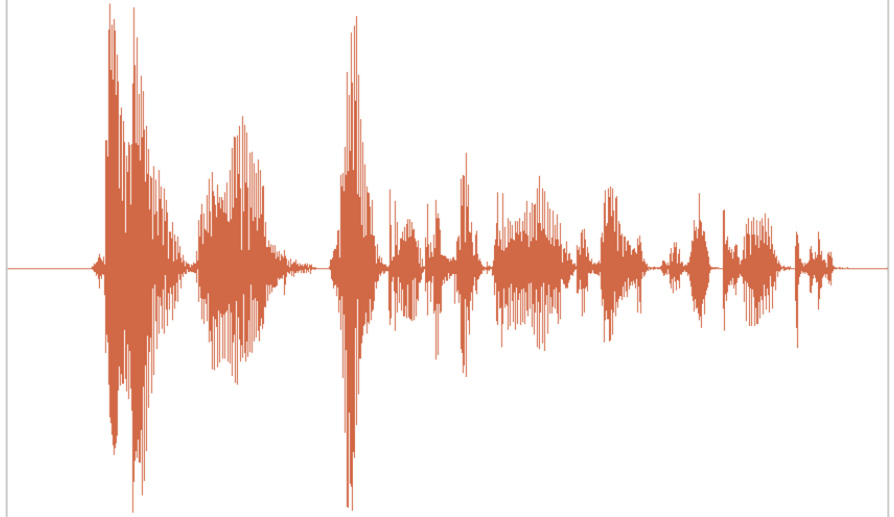


Source: Google Research – Tone Transfer

Speech Recognition Demo

Demo Overview

Input



Output

Hello World, welcome to the AI developer's toolkit.

Demo Options

Microsoft

Client SDK

Google

Client SDK

Amazon

Client SDK

Microsoft

REST API

Google

REST API

Speech

Cognitive Services

Search (Ctrl+/)

Overview
Activity log
Access control (IAM)
Tags
Diagnose and solve problems

RESOURCE MANAGEMENT

Keys and Endpoint
Encryption
Pricing tier
Networking
Identity
Billing By Subscription
Properties
Locks

Monitoring

Alerts
Metrics

Delete

Essentials

JSON View

Resource group (change)
Cognitive-Services

Status
Active

Location
East US

Subscription (change)
RenzeConsulting

Subscription ID

Tags (change)
Click here to add tags

API type
Speech

Pricing tier
Free

Endpoint
https://eastus.api.cognitive.microsoft.com/sts/v1.0/issuetoken

Manage keys
Click here to manage keys

Get Started | Discover | Develop | Deploy

Get Started with Speech Services

The Speech service is the unification of speech-to-text, text-to-speech, speaker recognition, speech translation, and intent recognition into a single Azure subscription.

Learn More

Discover

Develop

https://github.com/matthewrenze/ai-developers-toolkit/

Summary

Audio

Audio Analysis

Audio Synthesis

Demo